Stanford CS224v Course

Conversational Virtual Assistants with Deep Learning

Lecture 8

Structured and Unstructured Data Queries

Monica Lam & Shicheng Liu

Retrieval + Question Answering

1. **Matching a set of records with a set of constraints?** [L3, L4)
   Constraints-satisfaction based retrieval and match (SMT→DB)
2. **Short questions on free-text** (L5)
   RAG with embedding-based retrieval
3. **Databases and knowledge bases** (L6, L7)
   Complex queries in SQL, SPARQL (Select, aggregate, joins…)
4. **Hybrid: databases + free-text knowledge** (L8 - this lecture)

5. Systematic reviews across many documents (L9)

Example 1: Yelp: DB + Free text



Please find me a spicy Chinese restaurant.

Name TEXT	**Cuisine** TEXT[]	**Rating** NUM(2,1)	**...**	**Popular_dishes** FREE TEXT	**Reviews** FREE TEXT
Hummus Mediterranean Kitchen	mediterranean,halal, salad	4.0	...	Chicken Kebab Plate, Lamb Beef Gyro, Marinated Chicken Gyros, ...	| t l ; d r | This is your place if you're looking for a healthy and filling meal whether it's a quick pick-me-up or casual dining, ...
Penny Roma	italian, venues & event spaces	4.0	...	Cacio E Pepe, Agnolotti Dal Plin, Albacore Tartare, ...	My girlfriend was craving pasta on a Monday night. ... We were not expecting such an intimate and romantic dining experience. The restaurant was candle lit, modern, and perfect for a date night. ...

- Blue columns contain structured data
- Yellow columns are free text
- A lot of rows (thousands)

A lot of info is in free-text

Corpus does not fit in context windows

Hybrid Data

	Structured	Free Text
Wiki	Wikidata, Wikipedia tables	Wikipedia
Restaurants	Cuisines, opening hours, address, rating	Reviews, popular dishes
Products	Product ID, cost, ratings, sizes, physical dimensions, color	Descriptions, reviews
Courses	Class number, time offered, instructor, building, pre-requisites	Descriptions, reviews
Business processes	Receipts, statements	Regulations
Medical	Patient demographics Prescription	Diagnosis history

Quiz:
How would you handle the combo?



Hey! Can you recommend me an Italian restaurant with a romantic atmosphere?

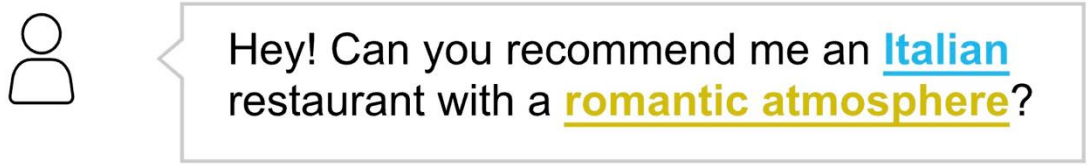
Method 1. Use a classifier to decide to use databases or text.

Does it work?

Jin et al., 2021; Zhao et al., 2023

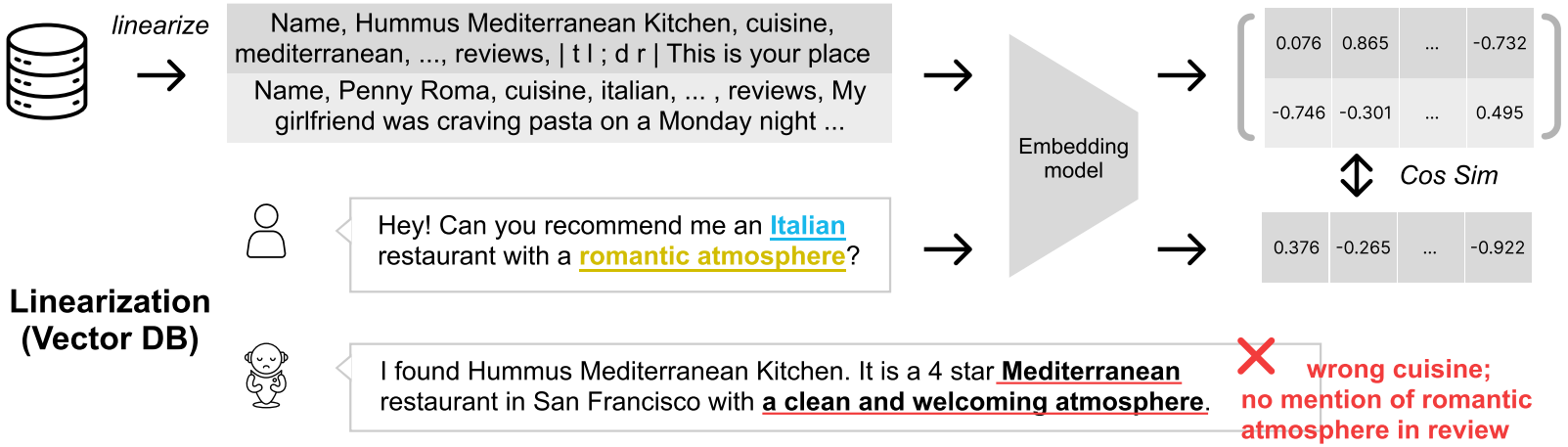
Method 2. Use the database to find the records, apply LLM to process text fields.

Jin et al. https://aclanthology.org/2021.dialdoc-1.16/
Zhao et al. https://aclanthology.org/2023.sigdial-1.28/



Method 3. Linearize structured data into free text

Oguz et al., 2022



Oguz et al., https://aclanthology.org/2022.findings-naacl.115/

Problems with Linearization

Do not take advantage of the power of SQL

- How to combine different clauses together?
  e.g. multiple clauses: "Are there any restaurants that have **a great view of the Golden Gate bridge** in **San Francisco**?"

- Hard to do column computations
  (rank, max, min, average, count) on flattened data
  - e.g. "what is the **top-rated** Chinese restaurant?"

HybridQA: An illustrative dataset

HybridQA: A Dataset of Multi-Hop Question Answering over Tabular and Textual Data

Wenhu Chen, Hanwen Zha, Zhiyu Chen, Wenhan Xiong, Hong Wang, William Wang
University of California, Santa Barbara, CA, USA
{wenhuchen, hwzha, xwhan}@cs.ucsb.edu,
{zhiyuchen, hongwang600, william}@cs.ucsb.edu

Findings of the Association for Computational Linguistics: EMNLP 2020, pages 1026–1036
November 16 - 20, 2020. ©2020 Association for Computational Linguistics

HybridQA Dataset (1 out of 13000 tables)

Wikipedia Tables *hyperlinked* Wikipedia Pages

Flag bearers of Myanmar at the Olympics

The 2016 Summer Olympics officially known as the Games of the XXXI Olympiad (Portuguese : Jogos da XXXI Olimpíada) and commonly known as **Rio** 2016 , was an international multi-sport event ……

Name	Year	Season	Flag bearer
XXXI	2016	Summer	Yan Naing Soe
XXX	2012	Summer	Zaw Win Thet
XXIX	2008	Summer	Phone Myint Tayzar
XXVIII	2004	Summer	Hla Win U
XXVII	2000	Summer	Maung Maung Nge
XX	1972	Summer	Win Maung

Yan Naing Soe (born **31 January 1979**) is a Burmese judoka . He competed at the 2016 Summer Olympics in the **men 's 100 kg event** , …… He was the flag bearer for Myanmar at the **Parade of Nations** .

Zaw Win Thet (born **1 March 1991** in Kyonpyaw , Pathein District , Ayeyarwady Division , Myanmar) is a Burmese runner who ……

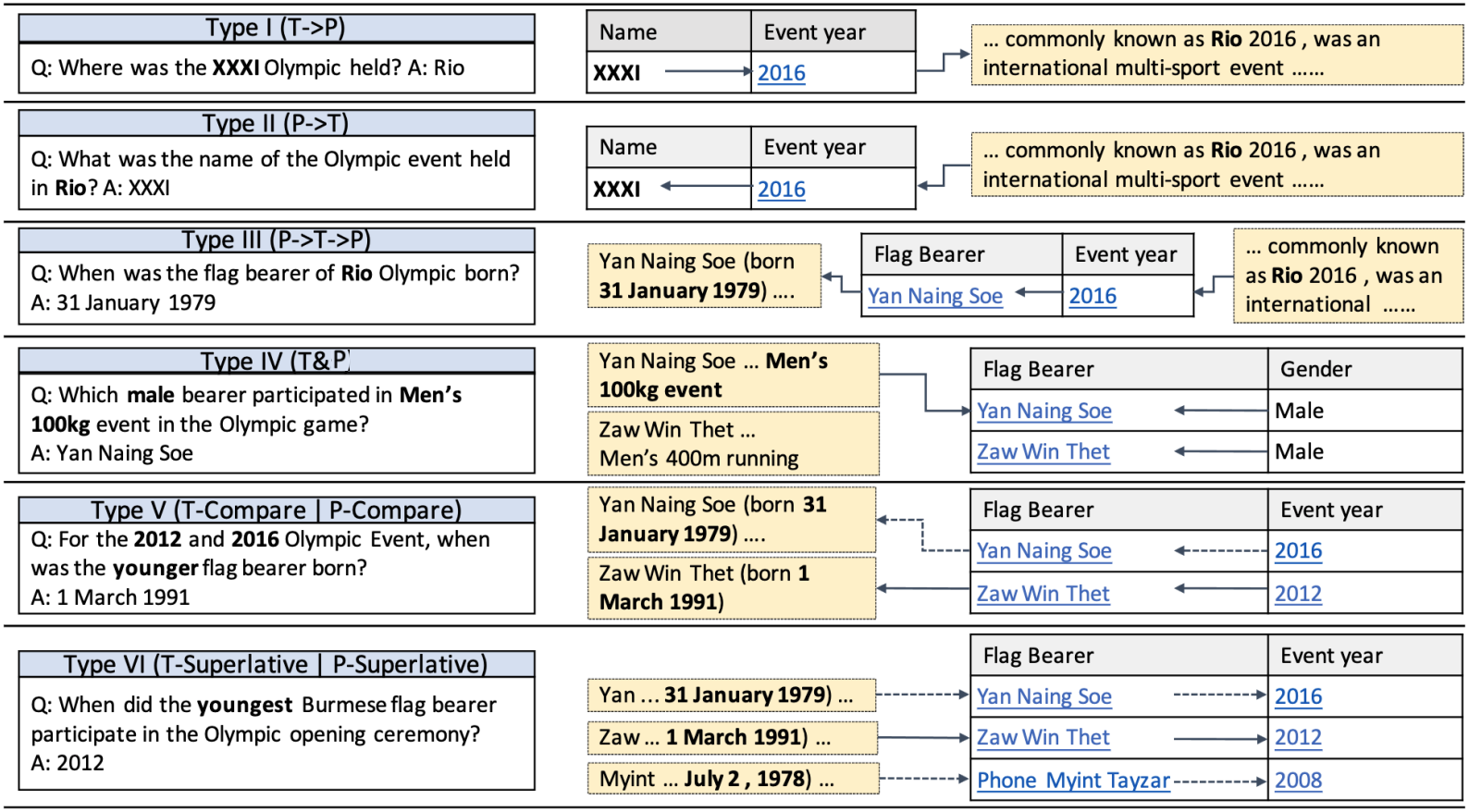
Myint Tayzar Phone (Burmese : မြင့်တေစောဖုန်း) born **July 2 , 1978**) is a sprint canoer from Myanmar who competed in the late 2000s .

……

Win Maung (born **12 May 1949**) is a Burmese footballer . He competed in the men 's tournament at the 1972 Summer Olympics …

Hardness

Q: In which year did the judoka bearer participate in the Olympic opening ceremony?	A: 2016
Q: Which event does the does the XXXI Olympic flag bearer participate in?	A: men's 100 kg event
Q: Where does the Burmesse jodoka participate in the Olympic opening ceremony as a flag bearer?	A: Rio
Q: For the Olympic event happening after 2014, what session does the Flag bearer participate?	A: Parade of Nations
Q: For the XXXI and XXX Olympic event, which has an older flag bearer?	A: XXXI
Q: When does the oldest flag Burmese bearer participate in the Olympic ceremony?	A: 1972



T: Table
P: Passage

Quiz: How many total # of types can there be?

Challenges

- Arbitrary composition of database and text accesses
  - Joins – combines information of columns and free-text
  - Superlatives – find max on entire columns

- Database queries are optimized for large datasets
  - Many many compiler optimizations e.g. order of joins, blocking the computation to improve locality, etc.
  - Any free-text search breaks up the queries, preventing optimizations across the whole query.

Previous techniques

Can these approaches handle the HybridQA questions (assuming large tables)?

1. Classify query to decide to retrieve from DB or Text
2. Retrieve answers from DB first and use LLM to process the text fields
3. Linearize DB to text; use Text Retrieval

Lecture Goals

- Understanding hybrid queries
- **SUQL Language – Design and Rationale**
- SUQL Implementation – Semantic parser & Compiler
- Evaluation in a real life app: Yelp
- Evaluation with a large dataset: HybridQA

Goals for Handling Hybrid Data

1. **Maximize available information retrieval capability**
   - Keep text as text and KBs as KBs
     - Combine the best of two worlds!
2. **Hybrid multi-hop: Support arbitrary composition of IR and KB**
   - Efficient searching of free text (using embedded similiarity)
   - Relational algebra operations (select, aggregate, rank, …) **NEW!**

SUQL: Conversational Search over Structured and Unstructured Data with Large Language Models

Shicheng Liu Jialiang Xu Wesley Tjangnaka
Sina J. Semnani Chen Jie Yu Monica S. Lam
Computer Science Department, Stanford University
Stanford, CA

- **Create the SUQL (Structured & Unstructured Query Language)**
  - Leverages LLM familiarity
    with the most popular DB query language
  - Extends SQL to include free-text operations
    - The language supports arbitrary composition
  - Create an optimizing compiler to evaluate efficiently

SUQL Free-Text Support

- Add free-text primitives into SQL, implemented with IR & LLM
- Two functions: summary, answer

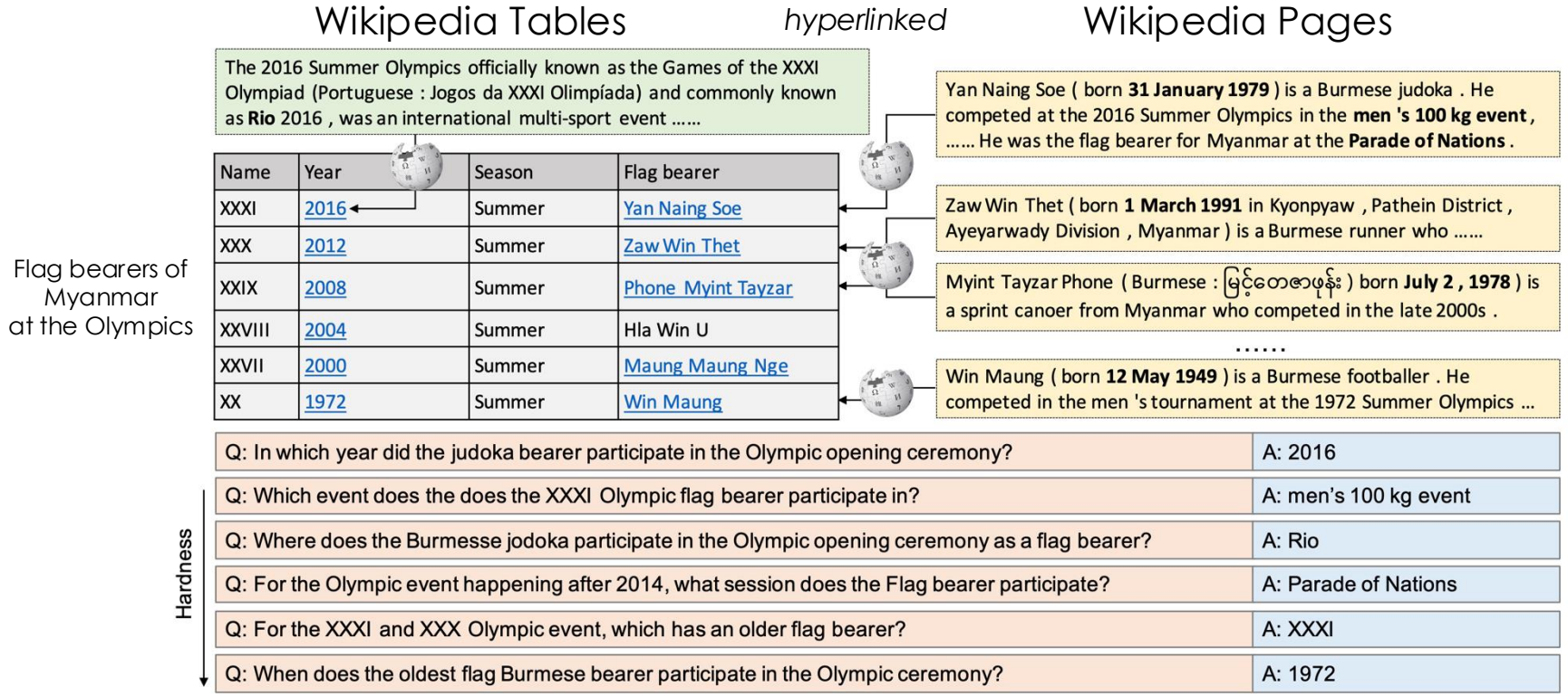
"I want a family-friendly restaurant in Palo Alto"

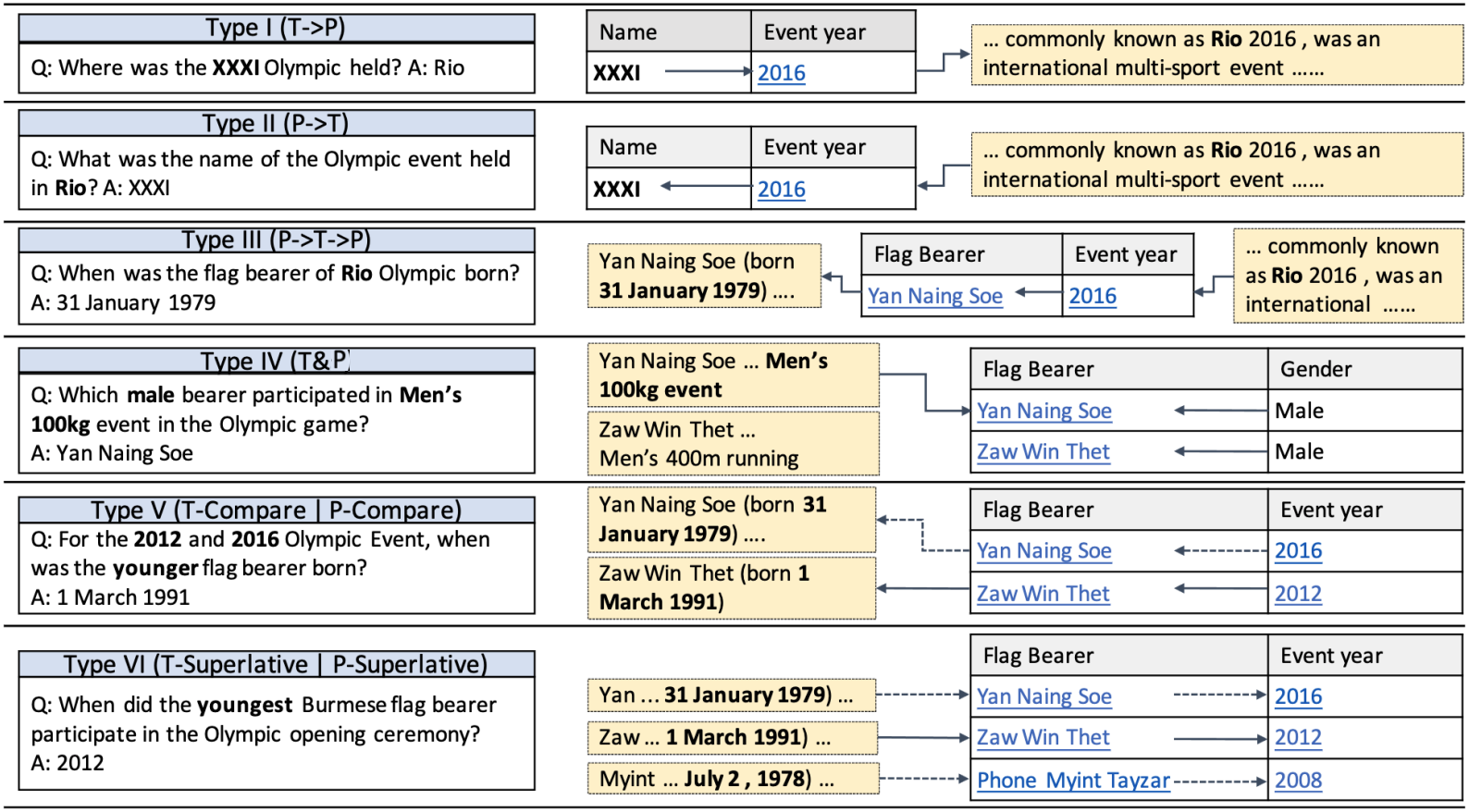


```
SELECT *, summary(reviews) FROM restaurants
    WHERE location = 'Palo Alto'
AND answer(reviews, 'is it a family friendly restaurant') = 'Yes' LIMIT 1;
```

Quiz: How do you implement summary and answer?

HybridQA Dataset





T: Table
P: Passage

Figure 3: Illustration of different types of multi-hop questions.

HybridQA Questions in SUQL

CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);

T: Table
P: Paragraph

Type I (T->P)

SELECT answer("Event year_Info",
'where is this event held?')
FROM "Flag" WHERE "Name" = 'XXXI'

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type I (T->P)

Where was the XXXI Olympic held?

```
SELECT answer("Event year_Info",
              'where is this event held?')
       FROM "Flag" WHERE "Name" = 'XXXI'
```

HybridQA Questions in SUQL

Type II (P->T)

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

```
SELECT "Name" FROM "Flag"
WHERE answer("Event year_Info",
        'where is this event held?') = 'Rio'
```

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type II (P->T)

What was the name of the Olympic event held in Rio?

```
SELECT "Name" FROM "Flag"
WHERE answer("Event year_Info",
        'where is this event held?') = 'Rio'
```

HybridQA Questions in SUQL

Type III (P->T->P)

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

```
SELECT answer("Flag Bearer_Info",
                    'when is this person born?')
FROM "Flag"
WHERE answer("Event year_Info", 'where is this event
held?') = 'Rio'
```

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type III (P->T->P)

When was the flag bearer of Rio Olympic born?

```
SELECT answer("Flag Bearer_Info",
                    'when is this person born?')
FROM "Flag"
WHERE answer("Event year_Info", 'where is this event
held?') = 'Rio'
```

HybridQA Questions in SUQL

Type IV (T&P)

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

```
SELECT "Flag Bearer" FROM "Flag" WHERE
"Gender" = 'Male' AND
answer("Flag Bearer_Info",
        'what event did this person participate in?')
        = "Men's 100kg event"
```

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type IV (T&P)

Which **male** bearer participated in **Men's 100kg** event in the Olympic game?

```
SELECT "Flag Bearer" FROM "Flag" WHERE
"Gender" = 'Male' AND
answer("Flag Bearer_Info",
        'what event did this person participate in?')
        = "Men's 100kg event"
```

HybridQA Questions in SUQL

Type V (T-Compare | P-Compare)

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

```
SELECT MAX
(answer("Flag Bearer_Info",
        'when is this person born?')::date)
FROM "Flag" WHERE "Event year" IN ('2016', '2012')
```

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type V (T-Compare | P-Compare)

For the **2012** and 2016 Olympic Event, when was the younger flag bearer born?

```
SELECT MAX
(answer("Flag Bearer_Info",
            'when is this person born?')::date)
FROM "Flag" WHERE "Event year" IN ('2016', '2012')
```

HybridQA Questions in SUQL

Type VI (T-Superlative | P-Superlative)

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

```
SELECT "Event year" FROM "Flag" ORDER BY
answer("Flag Bearer_Info",
        'when is this person born?')::date
DESC LIMIT 1;
```

HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

T: Table
P: Paragraph

Type VI (T-Superlative | P-Superlative)

When did the **youngest** Burmese flag bearer participate in the Olympic opening ceremony?

```
SELECT "Event year" FROM "Flag" ORDER BY
answer("Flag Bearer_Info",
        'when is this person born?')::date
DESC LIMIT 1;
```

At-A-Glance: HybridQA Questions in SUQL

```
CREATE TABLE Flag (
"Name" TEXT,
"Flag Bearer" TEXT,
"Flag Bearer_Info" TEXT[],
"Gender" TEXT,
"Event year" TEXT,
"Event year_Info" TEXT[]);
```

Type I (T->P)
Where was the XXXI Olympic held?

```
SELECT
answer("Event year_Info",
        'where is this event held?')
FROM "Flag" WHERE "Name" = 'XXXI'
```

Type II (P->T) What was the name of the Olympic event held in Rio?

```
SELECT "Name" FROM "Flag"
WHERE answer("Event year_Info",
'where is this event held?') = 'Rio'
```

Type III (P->T->P) When was the flag bearer of Rio Olympic born?

```
SELECT answer("Flag Bearer_Info",
'when is this person born?')
FROM "Flag"
WHERE answer("Event year_Info",
'where is this event held?') = 'Rio'
```

Type IV (T&P) Which **male** bearer participated in **Men's 100kg** event in the Olympic game?

```
SELECT "Flag Bearer" FROM "Flag"
WHERE "Gender" = 'Male' AND
answer("Flag Bearer_Info",
'what event did this person
participate in?')
= "Men's 100kg event"
```

Type V (T-Compare | P-Compare) For the **2012** and 2016 Olympic Event, when was the younger flag bearer born?

```
SELECT MAX
(answer("Flag Bearer_Info",
  'when is this person born?')::date)
FROM "Flag"
WHERE "Event year" IN ('2016', '2012')
```

Type VI (T-Superlative | P-Superlative) When did the **youngest** Burmese flag bearer participate in the Olympic opening ceremony?

```
SELECT "Event year" FROM "Flag"
ORDER BY answer("Flag Bearer_Info",
'when is this person born?')::date
DESC LIMIT 1;
```

Quiz: Where can 'answer' be used in a query?

Conversational Example: Restaurant

Restaurants

Do you have a recommendation for a first date restaurant in Palo Alto?
We're thinking sushi but not sure what's good around here.



```
SELECT *, summary(reviews) FROM restaurants
WHERE 'sushi' = ANY (cuisines) AND location = 'Palo Alto' AND rating >= 4.0
AND answer(reviews, 'is this restaurant good for a first date?') = 'Yes'
ORDER BY num_reviews DESC LIMIT 1;
```

Conversational Example: Shopping

```
CREATE TABLE laptop (
"ram" int,
…,
"about" TEXT,
“description" TEXT[],
"reviews" TEXT);
```

Laptops

I need a laptop with a Thunderbolt 3 port and at least 16GB RAM for my workstation setup.

↓

```
SELECT *, summary(reviews) FROM laptops
WHERE ram >= 16 AND
    (answer(about, 'does this laptop have Thunderbolt 3?') = 'Yes’
  OR answer(description, 'does this laptop have Thunderbolt 3?') = 'Yes’)
LIMIT 3;
```

It is often unclear where the answer is in the text, useful/easy to search for all possibly relevant text fields.

QUIZ

WHAT CAN WE USE SUQL FOR?

Advantages of SUQL

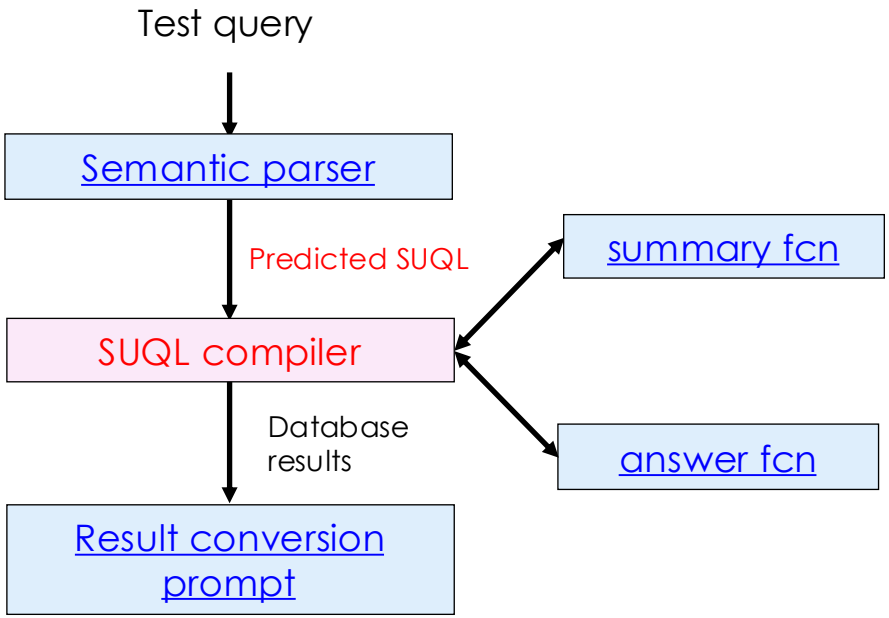
- **Formal representation combining IR and DB query**
  - Succinct, expressive, domain independence
  - Full compositionality: combining IR and relational algebra
- **Interpretability**
  - We can read the query back
    so users know what question is being answered
  - Quiz: Is the answer always correct given a formal query?

- **SUQL is a high-level programming language**
  - Optimization issues handled by compilers, abstracted from programmers

Lecture Goals

- Understanding hybrid queries
- SUQL Language – Design and Rationale
- **SUQL Implementation – Semantic parser & Compiler**
- Evaluation in a real life app: Yelp
- Evaluation with a large dataset: HybridQA

SUQL-based agent on HybridQA
A simple prompt-based system



Repeat if there are no results

- SQL can run SUQL programs without modification
  - summary / answer are just external functions
- But it is slow

Templates written in jinja syntax

SUQL Optimizing Compiler

Example

"I want a family-friendly restaurant in Palo Alto"


```
SELECT *, summary(reviews) FROM restaurants
WHERE answer(reviews, 'is it a family friendly restaurant') = 'Yes'
and location = 'Palo Alto'
```

- Consider a naïve implementation
  - Retrieve each review from the table
  - For each review, call LLM to answer the question
  - Filter on location
  - Summarize all the returned reviews
  - Report the first few

Quiz: How would you optimize the query?

1. Use embedding to implement filter

answer(reviews, ‘is it a family friendly restaurant’) = ‘Yes’

- Use embedding model (vector similarity) to return top candidates, inside faiss (Facebook AI Similarity Search)
- For applications such as recommendation, it is not necessary to return all the answers
- Return only top results to LLM-based answer functions

2. ENUM classification

WHERE '**coffee**' = ANY(cuisines)

- "cuisines" is an enumerated type.
- Here, database only contains
  - **'coffee & tea'**
  - **'café'**
- Overload **=** operator
- Use LLM to implement a ENUM classifier to match against similar values

3. Predicate Ordering

answer(reviews, ‘is it a family friendly restaurant’) = ‘Yes’
AND
and location = 'Palo Alto’

- Execution of structured predicates is much cheaper
- Always execute structured predicates first

4. Lazy Evaluation

```
answer(reviews, ‘is it a family friendly restaurant’) = ‘Yes’
AND
answer(reviews, ’is it easy to park’) = ‘Yes’
AND
and  location = 'Palo Alto’ LIMIT 1
```

- Lazy evaluation: Evaluate only when the result is needed
- Stop calling answer as soon as LIMIT 1 is reached

SUQL Compiler Implementation

- SUQL is implemented by modifying the SQL compiler
- To support arbitrary composition of SQL and summary/answer statements
  - Process the syntax tree of SQL recursively
    - Start from the bottom node with no sub-queries
    - And move to the top
  - For each SELECT statement with answer in it
    Apply optimizations to the SELECT statement
    - Store the processed results in a temporary table *temp*
    - Substitute this statement with *SELECT from temp*
- SQL compiler handles the final processing

Compiler Optimizations

Compiler optimization needs to operate across the entire query

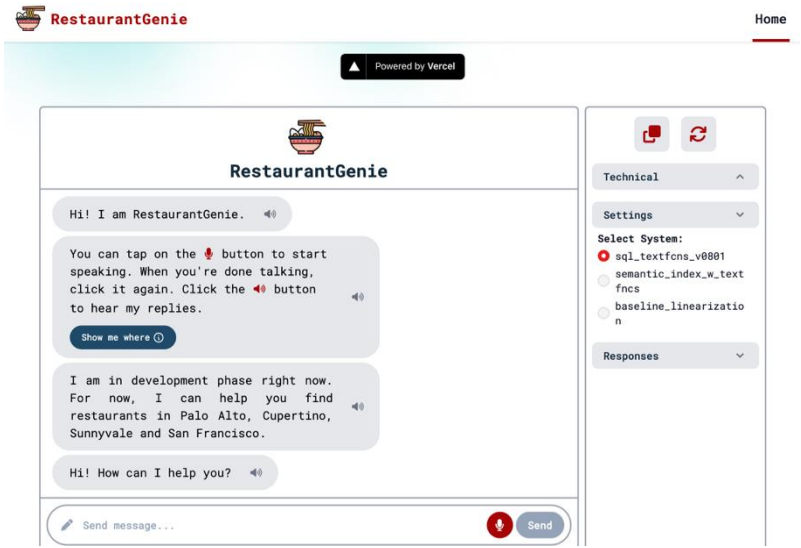
Quiz: Is the optimization necessary?

Quiz: What if we use an agent
that uses "function calling" to call IR
and query databases separately?

Lecture Goals

- Understanding hybrid queries
- SUQL Language – Design and Rationale
- SUQL Implementation – Semantic parser & Compiler
- **Evaluation in a real life app: Yelp**
- Evaluation with a large dataset: HybridQA

Experiment



A real, large dataset

- Yelp on San Francisco, Palo Alto, Cupertino, Sunnyvale
- Scraped reviews and popular dishes information

Components

- LLM: gpt-3.5-turbo
- Information retrieval on text fields
  - indexed with Coco-DR
- A new optimizing SUQL compiler

COCO-DR reference: https://arxiv.org/abs/2210.15212

Evaluation (in Restaurants)

- **We solicit user queries via crowdsourcing on Prolific.**
  - We do not disclose to the workers what fields are available in the database.

- **The Yelp Dataset:**
  1. Just Q&A: 100 crowdsourced questions about restaurants.
  2. Conversational: 20 conversations with 96 turns

The Need for Hybrid QA in Real Life?

	Single-turn	Conversation
Structured-only	45	37 (59%)
Combination	55	25 (41%)
Total	100	62

Evaluation (in Restaurants)

Quiz: how do you evaluate its accuracy?

- In real-world tasks: Difficult to label the gold answers:
  - For each query, there are many possible "gold restaurants"

- We manually inspect whether restaurants returned satisfy all criteria
- Calculate Query *precision*: #correct results / #results

Using SUQL



Hey! Can you recommend me an **Italian** restaurant with a **romantic atmosphere**?

↓ *Semantic Parser*

```
SELECT *, summary(reviews) FROM restaurants
WHERE 'italian' = ANY (cuisines) AND
answer(reviews, 'is this restaurant romantic?') = 'Yes'  LIMIT 1;
```

SUQL Compiler →





I found Penny Roma, which has a 4.0 rating on our database and offers a variety of **Italian** dishes. Overall, the atmosphere is described as delightful, authentic, and **perfect for a date spot**.



Evaluation

	Single-turn	Conversational
Linearization @ 1	57.0 %	63.4 %
Linearization @ 3	49.7 %	61.9 %
SUQL	**93.8 %**	**90.3 %**

Table 5: Turn accuracy measurement on linearized system versus SUQL system. (Linearization @ n: n answers)

Quiz: Why are there errors in SUQL answers?

Defensive Programming:
What if the Semantic Parser is Wrong?

Always verbalize the user query

Allows the user to spot mistakes in the answer

Even when it fails, so users can adjust the query.

I searched for 5-star restaurants in Sunnyvale that serve kids food.

Unfortunately, I couldn't find any search results.

Is there anything else I can help you with?

Recall Error Analysis

Hard to find all missing restaurants.
We performed analysis only when no answers are returned

14 false negatives on our single-turn set (100 questions)

- Parsing: 2 errors (one syntactic, one enum field confusion)
- Query evaluation ← Can be improved
  - Structured:
    - 4 unsupported location service
    - 2 opening hours errors
    - 2 dishes: searched in popular dishes, but results are in reviews
  - Free-text: 4 false negatives from 'answer' (ChatGPT)

Lecture Goals

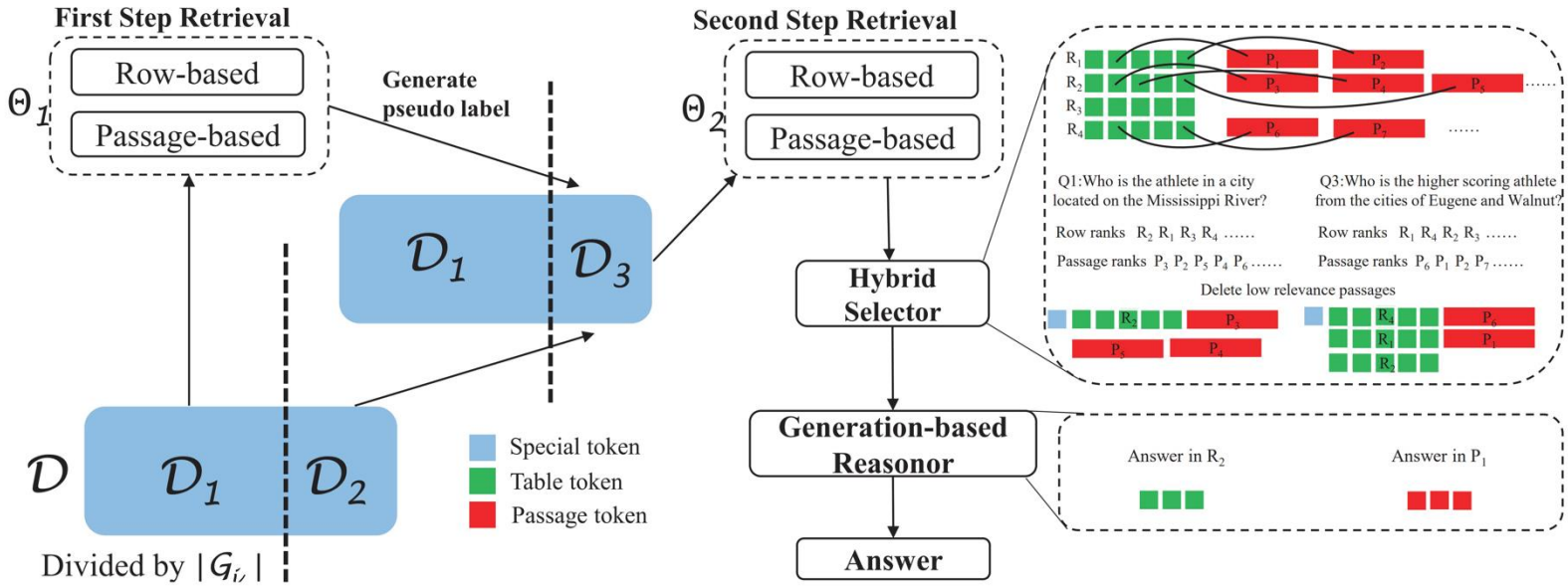
- Understanding hybrid queries
- SUQL Language – Design and Rationale
- SUQL Implementation – Semantic parser & Compiler
- Evaluation in a real life app: Yelp
- **Evaluation with a large dataset: HybridQA**

HybridQA Dataset: Based on Wikipedia

- 13000 tasks on Amazon Turk:
  - Given one crawled Wikipedia table + its hyperlinked passages.
  - Write 6 questions & answers.
  - Each task takes 12 minutes on average, and payment is $2.3 U.S.
- 5 CS graduate students to decide on the acceptance.

#Table	#Row/#Column	#Cell	#Links/Table
13,000	15.7/4.4	70	44
#Passage	#Words/Passage	#Ques	#Words/Ques
293,269	103	69,611	18.9

A Custom Retriever for HybridQA with model training on training set



Only uses LLM in the "generation-based reasoner" step

https://github.com/wenhuchen/HybridQA#recent-papers
SOTA model: https://arxiv.org/abs/2305.11725

Method	Model	Trained on (Size)	Dev		Test	
			EM	F1	EM	F1
DocHopper (Sun et al., 2022)	ETC	HybridQA (62k)	47.7	55.0	46.3	53.3
HYBRIDER (Chen et al., 2020)	BERT		44.0	50.7	43.8	50.6
MuGER[2] (Wang et al., 2022)			57.1	67.3	56.3	66.2
Mate (Eisenschlos et al., 2021)			63.4	71.0	62.8	70.2
DEHG (Feng et al., 2022)			65.2	**76.3**	63.9	75.5
MITQA (Kumar et al., 2023)			65.5	72.7	64.3	71.9
MAFiD (Lee et al., 2023b)	T5		66.2	74.1	65.4	73.6
S[3]HQA (Lei et al., 2023b)	BERT/BART/DeBERTa		**68.4**	75.3	**67.9**	**75.5**
LLaMA2 (7B) (Zhang et al., 2023a)	LLaMA2 (7B)	Zero-shot	20.7	-	-	-
TableLlama (Zhang et al., 2023a)		TableInstruct (2.6M)	27.6	-	-	-
End-to-End QA w/ retriever (Shi et al., 2024)	GPT-4	Zero-shot	24.5†	30.0†	-	-
HPROPRO (Shi et al., 2024)		Few-shot	48.0†	54.6†	48.7	57.7
SUQL (**Ours**)			59.3	68.3	59.0	68.4

Table 3: Performance of few-shot-based SUQL and related work on the HybridQA dataset. † denotes running on 200 sampled cases from the development set (Shi et al., 2024).

- SUQL outperforms all ICL methods and comes within 8.9% EM and 7.1% F1 to the SOTA trained on 62k examples
- None of the approaches on HybridQA **generalizes to large databases**

QUIZ

WHAT SHOULD WE DO NEXT?

Error Analysis

We analyzed 72 randomly sampled error cases

Evaluation Issues (61% of errors)

- **Format mismatches**: **37.5%**.
  Predicted "Johnson City, Tennessee"
  vs. "Johnson City".
- **The gold label is either wrong or incomplete**: **23.6%**
  - There is only 1 gold answer when there should be more!

True Errors (39% of errors)

- Semantic parsing errors: 22.2%
- Errors from LLM-based functions: 11.1%
- Type-related conversation errors: 5.6%

A lot of format problems ! ← Handled by fine-tuning, but it is not intrinsically important

True accuracy may reach 84.2%

Conclusion: SUQL

- **Hybrid data queries are natural and important**
  - Requires arbitrary composition of textual and DB retrieval
  - Requires query optimization to support arbitrary composition
- **Solution: SUQL**
  - Zero-Shot semantic parser: IR + relational algebra
  - An optimizing compiler
- **Evaluation**
  - Needed in real-life conversations (Yelp)
  - Handles composition of queries well (HybridQA)